CAPSTONE PROJECT

# MISSION YUVA

## Loan Analytics & Financial Inclusion Report

*Government of Jammu & Kashmir — Youth Entrepreneurship Initiative*

| | |
|---|---|
| TEAM ID | Group - 1 |
| INSTITUTE | Newton School of Technology, Rishihood University |
| SECTOR | Banking / MSME Finance / Govt Scheme Analytics |
| TOOL | Tableau |
| SUBMITTED | April 2026 |


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FACULTY MENTOR

*Archit Raj*


| 2,610 | 546 Cr | 4 | 188 Days |
|---|---|---|---|
| Applications | Total Loan | Dashboards | Avg TAT |

# Context & Problem Statement

Sector Background · Stakeholders · Objectives

### SECTOR CONTEXT

Mission Yuva is a J&K Government initiative providing collateral-free loans to youth entrepreneurs across 8 economic sectors in 5 districts: Doda, Pulwama, Reasi, Samba & Shopian. Borrowers operate in Agriculture, Healthcare, IT, Manufacturing, Retail, Services, Tourism, and Others.

### CORE BUSINESS QUESTION

**Which districts, sectors, and demographic segments carry elevated repayment risk — and where are the TAT bottlenecks preventing timely disbursement?**

### STAKEHOLDERS

- District SLBC Officers — approvals & TAT management
- J&K Mission Yuva Nodal Agency — policy & eligibility
- Partnering banks (SBDU, DLIC) — credit risk & disbursement
- Youth entrepreneurs aged 18–35 (rural & urban)

### OBJECTIVES

- Identify key drivers of late payment and default
- Quantify TAT bottlenecks across 5-stage disbursement pipeline
- Characterise applicant demographics for equitable allocation
- Build 4 interactive dashboards covering 11 KPIs

# Data Engineering

Source · Coverage · Cleaning · Data Dictionary

**SOURCE**

Mission Yuva Scheme
Administrative Database

**RECORDS**

**2,610**
Loan Applications

**DISTRICTS**

**5**
Doda · Pulwama · Reasi · Samba · Shopian

**SECTORS**

**8**
Agri · Health · IT · Mfg · Retail · Services · Tourism · Others

**MAJOR CLEANING STEPS**

- Standardised sector names (capitalisation & whitespace trimming)
- Converted currency fields to Cr / K display notations
- Derived binary flags for Late & Default repayment status
- Computed TAT per stage; flagged negative TAT outliers
- Created Risk Score bins (0-10, 10-20 … 90-100) for histograms
- Calculated EMI Ratio = Loan / (Income × 60) for affordability

**DATA DICTIONARY — KEY ATTRIBUTE GROUPS**

| | |
|---|---|
| • **Applicant Identity** | App ID, Name, Gender, DOB |
| • **Geography** | District, Residential Type |
| • **Loan Financials** | Loan Amount, Project Cost, Income |
| • **Risk & Credit** | Risk Score (0-100), Credit Score |
| • **Repayment** | Status: On-Time / Late / Default |
| • **TAT Pipeline** | 5-stage days: BHD→SBDU→DLIC→Bank→Disburse |
| • **Inclusion** | Specially Abled Type |

# KPI Framework

11 KPIs engineered across 4 dashboards

KPI-1
**Total Applications**
**2,610**

KPI-2
**Total Loan Disbursed**
**₹546 Cr**

KPI-3
**Avg Risk Score**
**51/100**

KPI-4
**Avg TAT**
**188 Days**

KPI-5
**Late Payments**
**863**

KPI-6
**Avg Credit Score**
**50/100**

KPI-7
**Total Defaulted**
**₹82 Cr**

KPI-8
**Female Applicants**
**2,610***

KPI-9
**Specially Abled**
**1,501**

KPI-10
**Rural Count**
**2,610***

KPI-11
**Avg EMI Ratio**
**0.30**

** KPI-8 and KPI-10 display 2,610 at the 'All' filter level; segment-specific counts appear after applying dimension filters.*

# Key EDA Insights

Observations across all 4 dashboards

### 📊 Loan Distribution

Portfolio concentrates in the ₹10–20 Lakh ticket-size range, peaking around ₹15 Lakh — aligned with MSME mandate. Drop-off is sharp beyond ₹25 Lakh.

### ⚠️ Repayment Stress

Only 34% (889) of applications are On-Time. 33% Late + 15% Default = 47.8% combined stress rate — far above healthy MSME NPA benchmarks.

### ⏱️ TAT Bottleneck

DLIC Approval is the longest and most variable stage across all 5 districts. Pulwama records the worst individual stage durations throughout the pipeline.

### 🗺️ District Disparity

Doda leads in both total and average loan amounts. Shopian shows comparable application volume but smaller ticket sizes, indicating lower project costs.

### 🌿 Sector Risk Spread

Tourism (~16%) and Agriculture (~15%) carry the highest default rates. Retail and Services perform comparatively better at ~10%. Risk scores are near-normally distributed (mean=51).

### 👥 Inclusion Signals

Rural applicants outnumber urban across all genders. Nano Entrepreneurs dominate enterprise types. 1,501 specially-abled applicants form a key inclusion segment.

# Advanced Analysis

Descriptive Segmentation · Cross-Tabulation · Distribution Analysis

**METHOD**

Descriptive Statistics · Cross-Tabulation Heatmaps · Histogram & Box Plot Distribution Analysis

**RATIONALE**

Objective is portfolio monitoring (not prediction); statistical segmentation reveals actionable risk pockets without over-fitting.

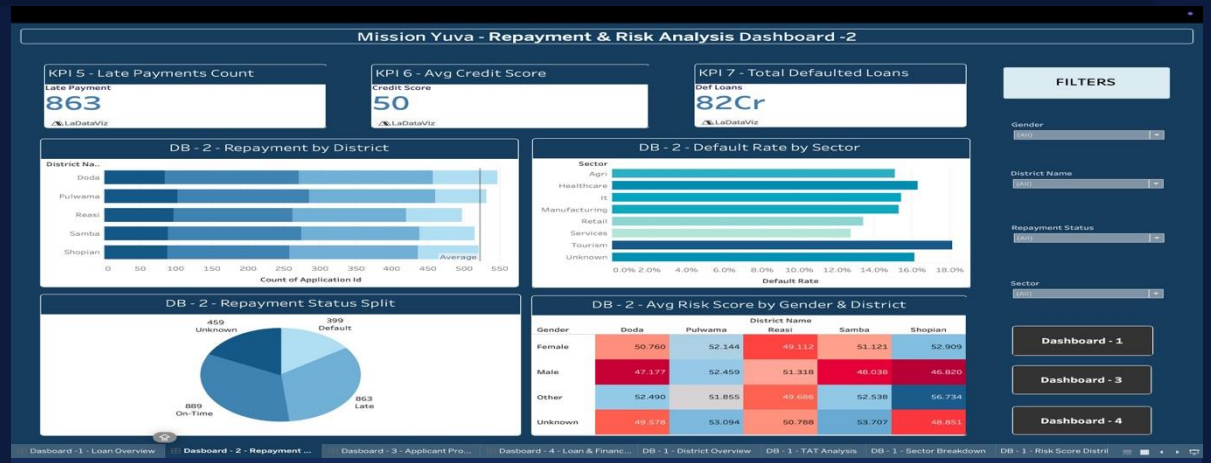
| ANALYSIS | RESULT | BUSINESS INTERPRETATION |
|---|---|---|
| **Default Rate by Sector** | Tourism 16.3% \| Agri 15.1% \| IT 13.2% \| Retail 10.4% | Tourism/Agri are highest-risk; Retail comparatively safer |
| **EMI Ratio Distribution** | Mean = 0.30; range ≈ 0.20–0.45 | Moderate leverage; <0.40 considered sustainable |
| **TAT Stage Analysis** | DLIC Approval is longest stage across all districts | Government approval is primary disbursement bottleneck |
| **Risk Score Spread** | Near-normal; mean = 51, SD ≈ 20 | Balanced risk portfolio; no extreme concentration |
| **Repayment Stress Rate** | (863 Late + 399 Default) / 2,610 = 47.8% | Nearly half the portfolio in some form of repayment stress |
| **Loan–Project Cost Correlation** | Near-perfect positive linear (R ≈ 0.99) | Scheme funds almost entire project cost — minimal borrower equity |

# Dashboard Overview

4 interactive dashboards · 11 KPIs · Multi-dimensional filters

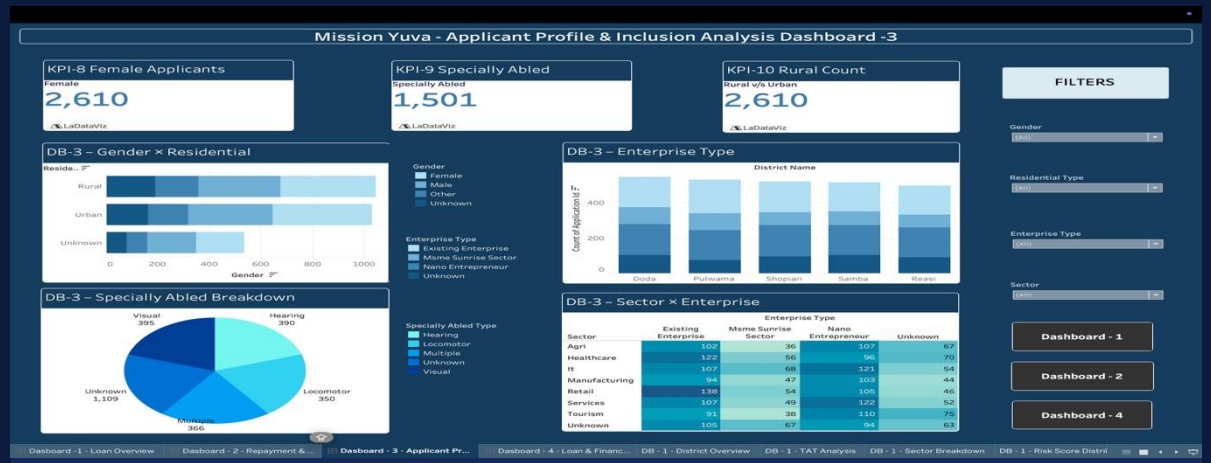
## DB-1: Loan Overview

KPI 1-4 · Sector heatmap · Risk distribution · TAT · District overview



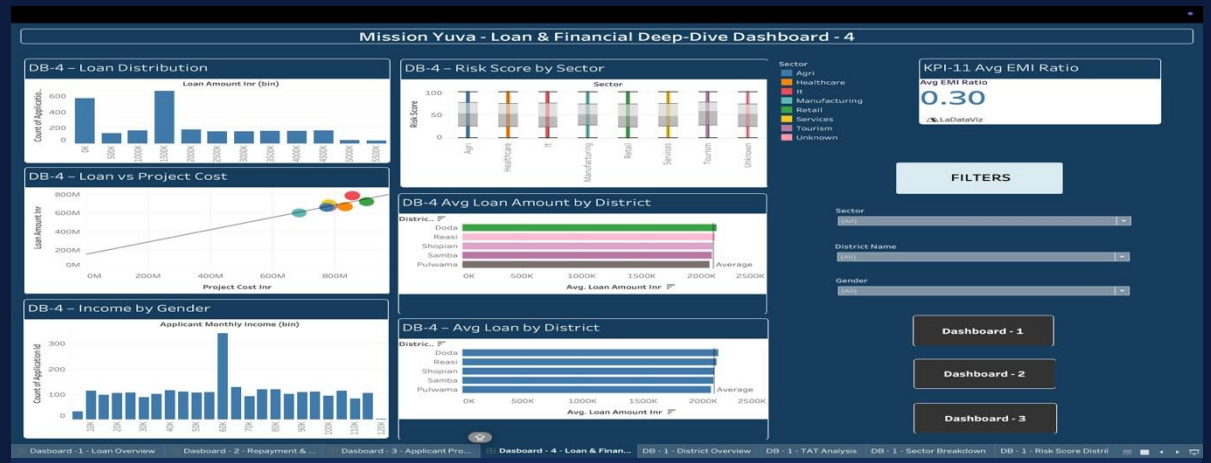

## DB-2: Repayment & Risk

KPI 5-7 · Default by sector · Risk by gender & district · Repayment split




## DB-3: Applicant Profile

KPI 8-10 · Gender × Residential · Enterprise type · Specially abled breakdown




## DB-4: Financial Deep-Dive

KPI-11 · Loan distribution · Loan vs Project cost · Avg loan by district

# Top Insights

12 high-value decision insights from the dashboards

### 01 Portfolio Stress Near-Critical

47.8% of portfolio (863 Late + 399 Default) is in repayment stress — far above healthy MSME benchmarks.

### 02 Tourism: Riskiest Sector

~16% default rate in Tourism. Post-pandemic demand disruption + seasonal J&K revenue volatility.

### 03 Agriculture: Structural Risk

15% default. Weather/market dependency without crop insurance or flexible repayment design.

### 04 DLIC Approval Bottleneck

Single biggest TAT driver across all districts. Streamlining it could halve the 188-day average turnaround.

### 05 Doda Dominates Loan Volume

Highest total & average loan amounts; suggests higher project costs or stronger admin capacity.

### 06 EMI Ratio Sustainable (0.30)

Moderate leverage; income peak at ₹50-70K/month means larger borrowers risk being over-leveraged.

### 07 Near-Full Project Funding

Loan–Project Cost correlation R≈0.99. Minimal borrower equity — increases portfolio risk exposure.

### 08 Female Rural Majority Segment

Rural female applicants form the largest demographic cohort — key inclusion success signal.

### 09 Risk Scores Poorly Differentiated

Near-normal distribution (mean=51) across all sectors; limited predictive segmentation power.

### 10 Specially Abled Underserved

1,109 'Unknown' out of 1,501 limits disability-disaggregated policy targeting.

### 11 Nano Entrepreneurs Dominate

Largest enterprise type across all 5 districts — scheme skews toward micro-scale borrowers.

### 12 Pulwama Highest Per-Stage TAT

Consistently worst individual stage durations; needs priority process intervention.

# Recommendations

5 actionable business recommendations for Mission Yuva

**R1**

### Sector-Differentiated Risk Underwriting

Apply tighter credit criteria and lower LTV ratios for Tourism and Agriculture loans. Introduce crop insurance linkage and seasonal repayment structures for Agri borrowers.

**R2**

### Digitise & Fast-Track DLIC Approval

Migrate DLIC approval workflow to a digital portal with SLA timers. Target ≤30-day DLIC turnaround to reduce the 188-day average TAT by an estimated 40%.

**R3**

### Mandate Minimum Borrower Equity Contribution

Require borrowers to fund at least 10-15% of project cost. The near-1:1 Loan:Project correlation signals zero equity skin-in-the-game — increasing moral hazard.

**R4**

### Repayment-Risk Early Warning System

Build a monthly monitoring flag for accounts crossing 60 days overdue. Trigger field-level credit counselling before accounts transition from Late to Default.

**R5**

### Improve Inclusion Data Quality at Application Stage

Mandate collection of Specially Abled type and Residential classification. 1,109 'Unknown' disability records and significant 'Unknown' residential entries limit policy targeting.

# Impact & Feasibility

Expected outcomes · Priority matrix

- Digitise DLIC approval → ~40% TAT reduction
- Early warning system → reduces Default rate
- Data quality mandates → better policy targeting

**HIGH IMPACT**
**HIGH FEASIBILITY**

- Sector-differentiated underwriting → lowers NPA
- Crop insurance linkage for Agriculture borrowers
- Equity contribution mandate → reduces moral hazard

**HIGH IMPACT**
**MEDIUM FEASIBILITY**

- Stress rate: 47.8% → target <25% in 2 years
- Avg TAT: 188 days → target <110 days
- Default rate: 15% → target <8% with early warning

**EXPECTED PORTFOLIO OUTCOMES**

# Limitations

Data constraints · Method boundaries

## DATA LIMITATIONS

- Gender field contains 'Unknown' category — not imputed, treated as separate segment
- Residential Type has 'Unknown' records — retained for completeness but limits urban/rural split accuracy
- 1,109 'Unknown' entries in Specially Abled Type — severely limits disability-disaggregated analysis
- Risk Score and Credit Score both average ~50-51 — suggests normalised scoring method with low predictive granularity
- TAT stage days are calendar days, not business days — overstates processing time
- Historical cohort only (no time-series or panel data) — cannot assess trend or seasonality
- Individual borrower characteristics (age, education, collateral) not captured in current dataset

## METHOD LIMITATIONS

- Descriptive analysis only — no predictive or causal inference model built
- Cross-tabulation heatmaps show association, not causation between variables
- Macro-economic forecasting excluded from scope — external shocks not accounted for
- Default Rate computed as count-based proportion — does not weight by loan size or exposure at default
- EMI Ratio assumes uniform 60-month repayment horizon for all borrowers — may overstate or understate affordability
- Box plots reveal IQR spread but not underlying multimodal distributions within sectors
- No statistical significance testing performed — patterns are directional, not validated

NEXT STEPS

# Future Extensions

→ Build a predictive default scoring model using ML (logistic regression / gradient boosting)

→ Integrate real-time repayment data feeds for live dashboard refresh

→ Conduct field surveys to capture qualitative borrower capacity indicators

→ Expand scope to remaining J&K districts and additional Mission Yuva cohorts

→ Develop borrower segmentation profiles for targeted credit counselling

→ Commission geo-spatial mapping of loan density and default concentration by sub-district

PROJECT SUMMARY

Mission Yuva loan analytics uncovered a portfolio at critical stress — with 47.8% of borrowers in late or default status, a 188-day end-to-end turnaround bottlenecked at DLIC approval, and Tourism & Agriculture carrying disproportionate credit risk.

Four dashboards, 11 KPIs, and 12 decision-ready insights equip the J&K lending authority with the visibility needed to intervene proactively, allocate equitably, and protect the scheme's revolving fund for future cohorts.

| 2,610 | ₹546 Cr | 47.8% | 188 Days |
|---|---|---|---|
| Applications | Portfolio | Stress Rate | Avg TAT |

# Thank You

Newton School of Technology · Rishihood University · April 2026